Article

Generalized Beam Theory with Exact Periodic Boundary Conditions for the Buckling Analysis of Omega-Stringer-Stiffened Composite Panels

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Abstract: Stringer-stiffened composite panels are commonly analysed as one representative cell of a wide panel with many identical stringers, and the side edges of that cell must then satisfy periodic boundary conditions. Closed-form models with periodic conditions rely on idealisations of the skin–stringer interaction, and shell finite element models with periodic constraints are accurate but expensive and give no modal insight. Generalized Beam Theory (GBT) offers a mechanically transparent and efficient description of thin-walled members, but its cross-section analysis has been developed for members with free longitudinal edges or with closed contours, and periodic side edges have not been treated. This study formulates a shear-deformable GBT for the omega-stringer-stiffened composite panel in which the periodic condition is built into every family of deformation modes: the conventional (global, distortional, local and transverse-extension) modes by constraint rows that are projected out family by family, the warping functions by a constrained Vlasov least-squares problem, and the shear and transverse-shear (Mindlin) modes by merging the nodal functions of the two side edges. Any combination of modes, and hence any truncated basis, therefore satisfies the periodic condition exactly. The walls are laminated Reissner–Mindlin plates with full membrane–bending and anisotropic coupling, the stringer is tied to the skin with its true offset, the pre-buckling state is the first-order solution under the actual end loads, and the amplitude functions are cubic B-splines in a Ritz sense, which covers simply supported and clamped loaded edges. The formulation reduces exactly to the classical GBT system for uncoupled orthotropic walls under uniform compression. Against 90 Abaqus shell models (five lay-ups, two stringer pitches, three aspect ratios) the critical loads agree within 0.58 % on average (maximum 1.48 %), and within 0.57 % (maximum 1.25 %) for the 30 cases with periodic side edges; the change caused by periodic instead of symmetry side edges is reproduced with a correlation of 0.995. For panels with bending–twisting or membrane–bending coupling, periodic side edges raise the critical load by up to about 2 % for simply supported and by 5–18 % for clamped loaded edges, the latter confirmed by an independent spline finite strip model.

Keywords: generalized beam theory; periodic boundary conditions; stiffened panels; omega stringer; laminated composites; buckling; shear deformation

1. Introduction

Stringer-stiffened panels are the standard construction of aircraft fuselage and wing covers, and their buckling behaviour is often the governing design criterion of weight-critical laminated composite structures [1,2]. Among the stiffener shapes, the omega (hat) stringer is attractive because its closed cell provides high bending and torsional stiffness, a wide bonding area and good damage tolerance [3,4]. A typical cover carries many identical stringers. Its buckling behaviour away from the free or supported longitudinal edges is therefore represented by one repeating cell, provided that the side edges of this cell are given the conditions that the neighbouring cells impose, namely periodic boundary conditions (PBC) [5–7].

1.1. Periodic boundary conditions in the analysis of stiffened panels

The treatment of the repeating cell is a classical topic of the analysis of stiffened panels. In exact and finite strip methods for prismatic plate assemblies, the infinitely wide panel was analysed early through its repeating portion: the exact strip programs VIPASA and VICONOPT treat infinitely wide or long panels by repeating a portion and its supports [8,9], and the finite strip method in general handles such assemblies with a small number of strips [10,11]. The underlying principle is the theory of waves in periodic structures (Bloch–Floquet theory), by which the response of an infinite periodic structure follows from one cell with phase-shifted boundary conditions [12].

At the Institute for Lightweight Engineering and Structural Mechanics (LSM), a series of closed-form and semi-analytical models for stringer-stiffened composite panels with periodic boundary conditions has been developed. Mittelstedt derived explicit buckling loads of blade-stringer-stiffened plates considering periodic boundary conditions [5] and extended the approach to the stiffener–plate interaction in local buckling [6]. For omega stringers, Mittelstedt and Beerhorst presented a closed-form buckling analysis [3], and Mittelstedt and Schröder treated the local postbuckling of hat-stringer-stiffened plates [13]. Schilling and Mittelstedt developed a closed-form approximate solution for the local buckling of omega-stringer-stiffened panels [4], a column-buckling model with periodic boundary conditions [7], and models for shear buckling, compression–shear interaction and local postbuckling [14–16]. Most recently, El Yaakoubi-Mesbah and Mittelstedt presented a closed-form analytical solution for the local buckling of omega-stringer-stiffened composite panels under compression [17], a computational model for local buckling [18] and a closed-form postbuckling analysis including the stringer-foot effects [19]; the stability of stiffened laminates is summarised in [20]. These models are efficient and transparent, but they rest on idealisations: the stringer is replaced by elastic restraints, the skin is reduced to one or a few plate strips with assumed shape functions, local and global buckling are usually treated by separate models, and the pre-buckling stress is taken as uniform. Membrane–bending coupling of unsymmetric laminates is typically included through a reduced bending stiffness [21].

The physics-informed neural network of the authors for the local buckling of omega-stiffened panels [22] used such a reduced one-dimensional energy model and three-dimensional Abaqus analyses as the reference. Shell finite element models with periodic constraint equations between the nodes of the two side edges are accurate and general, but each design requires a new mesh, a large eigenvalue analysis and, in the case of coupled laminates, care in the definition of the constraint set; they also provide the buckling mode only as a nodal field, without a decomposition into mechanically meaningful deformation mechanisms.

1.2. Generalized Beam Theory

Generalized Beam Theory (GBT), originated by Schardt [23,24] and developed further by Davies and Leach [25], describes the deformation of a prismatic thin-walled member as a superposition of cross-section deformation modes whose amplitudes vary along the member axis. A cross-section analysis determines the modes and the associated modal matrices, and a member analysis determines the amplitude functions. The modal decomposition separates global, distortional, local and shear-related deformations and has made GBT one of the most transparent tools for coupled instabilities of thin-walled members, as summarised in the state-of-the-art reports by Camotim et al. [26] and Mittelstedt [27,28]. Silvestre and Camotim formulated first- and second-order GBT for arbitrary orthotropic materials [29,30], and later shear-deformable GBT for composite members and its vibration counterpart [31,32]. Dinis et al. extended the cross-section analysis to arbitrarily branched open sections [33]. Gonçalves et al. introduced a new approach to the calculation of the deformation modes [34], and Bebiano et al. rationalised the cross-section analysis into natural, shear and transverse-extension mode families [35], together with the shear deformation modes [36], implemented in the code GBTUL 2.0 [37]. Closed sections were treated, among others, for circular cylindrical shells and tubes [38] and for regular polygonal tubes [39], where the closed contour leads to Fourier-type modes along the circumference. For laminated composites, GBT has been applied to FRP open-section columns [40] and to laminated CFRP cylindrical stiffened panels [41], and Taig and Ranzi formulated GBT for stiffened sections [42].

Kharghani and Mittelstedt combined GBT with the Ritz method into a GBT–Ritz approach: for global, local and distortional buckling of shear-deformable composite beams with open cross-sections [43], for composite box girders [44], and for thin-walled composite beams with unsymmetric layups, for which membrane–bending coupling is consistently included [45,46]. In the GBT–Ritz approach, GBT supplies the cross-section kinematics and the modal matrices, and the Ritz method supplies the approximation of the amplitude functions along the member, with admissible functions that satisfy the essential boundary conditions [47,48]. The present work follows this route.

1.3. Research gaps and motivation

Two gaps motivate the present study.

The gap in GBT. GBT has been developed for members: open sections with free longitudinal edges, branched sections, and closed contours. The skin of a stiffened panel cell, however, ends at two side edges that are neither free nor closed: the displacement field at one side edge must equal that at the other. This condition has not been formulated in GBT. It cannot simply be imposed afterwards on a set of classical modes. Clas-

sical mode generation starts from a few natural nodes, and the condition couples the two ends of the open contour; a truncated set of classical modes then contains too few modes that satisfy it, and the rigid-body and warping modes of the open cell are not admissible for the periodic cell. Moreover, the omega stringer forms closed cells with the skin, so that Vlasov's null membrane shear condition cannot be fulfilled by the warping functions, the stringer is offset from the skin by the half thicknesses, and the pre-buckling state is not uniform near the loaded edges, where the load enters the stringer through the eccentric foot.

The gap in the treatment of periodic conditions. The closed-form models with periodic conditions summarised above are efficient, but they rely on idealisations of the skin–stringer interaction, separate local and global buckling, and do not account for the non-uniform pre-buckling state. Shell finite element models with periodic constraints do not have these restrictions but are expensive for parametric studies and give no modal insight. Exact and finite strip methods treat the repeating portion efficiently, but they are formulated in nodal degrees of freedom and do not provide the decomposition into global, distortional and local mechanisms.

Why combine them. GBT and periodic boundary conditions complement each other. The periodic cell is prismatic, so the member analysis of GBT applies directly; the modal decomposition identifies the buckling mechanisms of the cell; and the reduced number of generalised coordinates makes parametric studies cheap. Conversely, the periodic condition removes the arbitrariness of the side-edge modelling that otherwise dominates panel models built from members. The key requirement is that the condition is satisfied by the modes themselves, so that it holds for any combination and any truncation of the basis.

1.4. Contributions

The contributions of this study are:

1. A cross-section analysis in which the periodic condition is built into every family of GBT deformation modes: conventional modes (global, distortional, local, transverse extension) through constraint rows that are projected out family by family, the warping of the global and distortional modes through a Vlasov least-squares problem with a periodicity constraint, and the shear and transverse-shear modes through merging of the nodal functions of the two side edges (Section 3.4). Every mode, and therefore every truncated basis, satisfies the periodic condition exactly.
2. A shear-deformable GBT with laminated Reissner–Mindlin walls including all membrane–bending and anisotropic couplings, transverse-extension and shear modes, a stringer–skin tie with the true offset, a first-order pre-buckling state under the actual end loads with all membrane resultants, and amplitude functions approximated by cubic B-splines in a Ritz sense, covering simply supported (SS) and clamped (CC) loaded edges (Sections 3.2–3.8).
3. The exact reduction of the formulation to the classical GBT system and to the classical geometric matrices for uncoupled orthotropic walls under uniform compression (Section 3.9).
4. A validation against 90 Abaqus shell models with symmetry and periodic side edges and an independent spline finite strip model, including the first systematic quantification of the effect of periodic instead of symmetry side edges on omega-stiffened panels with coupled laminates (Section 4).

The paper is organised as follows. Section 2 defines the panel and the load. Section 3 develops the formulation: cross-section discretisation, constitutive law, kinematics, deformation modes, periodic conditions, energy and modal matrices, amplitude functions, pre-buckling state and eigenvalue problem, and the classical limits. Section 4 presents the verification, and Section 5 the conclusions.

2. Problem definition

2.1. Geometry and coordinates

The panel is the periodic cell of an infinitely wide panel with identical omega stringers at the pitch b_s (Figure 1a). The cell contains two stringers and has the width $W = 2b_s$ and the length $L = \beta b_s$, with the aspect ratio β. The global axes are X across the panel, Y normal to the skin towards the stringers, and x along the stringers, which is the load direction and the longitudinal axis of GBT. The skin occupies $-W \leq X \leq 0$, $Y = 0$, $0 \leq x \leq L$; the stringer centre lines are at $X = -b_s/2$ and $X = -3b_s/2$, so that the side edges $X = -W$ and $X = 0$ lie in the middle of a skin bay.

The stringer geometry is fixed: web root half-width $b_3 = 27.75$ mm, web height $h = 30$ mm, web inclination $\theta = 25^\circ$ from the vertical, crown half-width $h_t = b_3 - h\tan\theta = 13.76$ mm, and foot (flange) width $b_1 = 0.1\,(b_s - 2b_3)$. The skin thickness is t_s and the stringer thickness t_f. The stringer mid-surface is offset from the skin mid-surface by $c = (t_s + t_f)/2$, so that the web roots lie at $Y = c$ and the crown at $Y = h + c$.

In each wall, s is the coordinate along the wall mid-line, z the coordinate through the thickness, and u, v, w are the displacements in the directions x, s and z, as in classical GBT (Figure 1b).

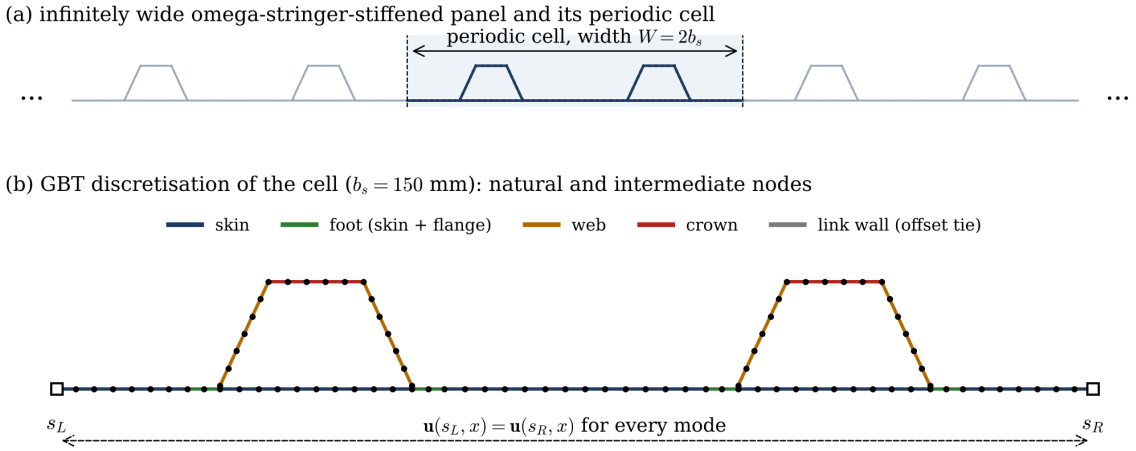


Figure 1. (a) Infinitely wide omega-stringer-stiffened panel and its periodic cell of width $W = 2b_s$. (b) GBT discretisation of the cell for $b_s = 150$ mm with natural and intermediate nodes; skin, feet (skin and flange as one laminate), webs, crowns and link walls of the offset tie; s_L and s_R are the side nodes on which the periodic condition acts.

2.2. Materials, lay-ups and load

Two materials are considered: an isotropic material ($E = 200$ GPa, $\nu = 0.3$, $t_s = 1.0$ mm, $t_f = 0.875$ mm) and a carbon-fibre reinforced plastic (CFRP) ply with $E_1 = 150$ GPa, $E_2 = 9$ GPa, $G_{12} = G_{13} = G_{23} = 5.3$ GPa, $\nu_{12} = 0.33$ and the ply thickness $t_{ply} = 0.125$ mm. The skin has eight and the stringer seven plies (Table 1). The lay-ups include a cross-ply laminate without coupling, two symmetric angle-ply laminates with bending–twisting coupling ($D_{16}, D_{26} \neq 0$, and in-plane shear–extension coupling $A_{16}, A_{26} \neq 0$ where the laminate is not balanced), and an unsymmetric laminate with membrane–bending coupling ($\mathbf{B} \neq \mathbf{0}$). Ply angles are measured from the stringer axis x.

Table 1. Lay-ups of skin (8 plies) and stringer (7 plies), ply thickness 0.125 mm; ply angles in degrees from the x-axis, subscript s: symmetric, subscript 2: repeated twice.

Name	Skin	Stringer	Coupling
Cross-ply	$[0/90/0/90]_s$	$[0/90/0/90/0/90/0]$	none
Sym1	$[45/-45/-45/45]_s$	$[45/-45/45/-45/45/-45/45]$	D_{16}, D_{26} (stringer also A_{16}, A_{26})
Sym2	$[60/-30/-30/60]_s$	$[60/-30/60/-30/60/-30/60]$	$A_{16}, A_{26}, D_{16}, D_{26}$
Unsym	$[0/30/60/90]_2$	$[0/30/60/90/0/30/60]$	**A**, **B**, **D** fully populated

A uniform compressive line load of 1 N/mm acts on every loaded edge at $x = 0$ and $x = L$: on the skin, the webs and the crowns, and 2 N/mm on a foot, which carries skin and flange; the foot load acts at the height of its resultant, $Y = c/2$. The eigenvalue λ is therefore the critical line load $N_{cr} = \lambda \times 1$ N/mm, the quantity reported by the Abaqus buckling step. The design points are $b_s \in \{150, 285.5\}$ mm and $\beta \in \{0.75, 1.5, 3\}$, with SS or CC loaded edges and symmetry or periodic side edges.

3. Theory and formulation

The formulation follows the two stages of GBT [24,29,35]: a cross-section analysis, which defines the deformation modes and the modal matrices of the periodic cell, and a member analysis, which determines the amplitude functions along x by the Ritz method, as in the GBT–Ritz approach [43,46]. The notation of classical GBT is used throughout: $\bar{u}_k(s)$, $\bar{v}_k(s)$, $\bar{w}_k(s)$ are the displacement profiles of mode k and $\varphi_k(x)$ its amplitude function.

3.1. Wall constitutive law

Every wall is a laminated Reissner–Mindlin plate [49–51]. With the generalised strains and stress resultants

$$\begin{aligned}\hat{\varepsilon} &= [\varepsilon_{xx},\ \varepsilon_{ss},\ \gamma_{xs},\ \kappa_{xx},\ \kappa_{ss},\ \kappa_{xs},\ \gamma_{xz},\ \gamma_{sz}]^{\mathsf{T}},\\ \hat{\sigma} &= [N_{xx},\ N_{ss},\ N_{xs},\ M_{xx},\ M_{ss},\ M_{xs},\ Q_x,\ Q_s]^{\mathsf{T}}\end{aligned} \tag{1}$$

the constitutive law is $\hat{\sigma} = \mathbf{C}\,\hat{\varepsilon}$ with

$$\begin{aligned}\mathbf{C} &= \begin{bmatrix}\mathbf{A} & \mathbf{B} & \mathbf{0}\\ \mathbf{B} & \mathbf{D} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{A}_s\end{bmatrix},\\ \left(A_{ij}, B_{ij}, D_{ij}\right) &= \sum_{p=1}^{n_l}\int_{z_p}^{z_{p+1}} \bar{Q}_{ij}^{(p)}\left(1, z, z^2\right) dz,\\ \mathbf{A}_s &= \tfrac{5}{6}\sum_{p=1}^{n_l}\bar{\mathbf{Q}}_s^{(p)}\left(z_{p+1}-z_p\right)\end{aligned} \tag{2}$$

where $\bar{\mathbf{Q}}^{(p)}$ and $\bar{\mathbf{Q}}_s^{(p)}$ are the transformed reduced in-plane and transverse shear stiffness matrices of ply p, $i, j \in \{1, 2, 6\}$ and n_l is the number of plies. All entries of **A**, **B** and **D** are retained, including the membrane–bending coupling **B** of unsymmetric laminates and the anisotropic terms $A_{16}, A_{26}, D_{16}, D_{26}$. The transverse membrane resultant N_{ss} is not condensed out, because the transverse-extension modes of Section 3.3 carry ε_{ss} explicitly.

A stringer foot consists of the skin and the stringer flange, which are tied in the shell model. In GBT the two layers form one wall referred to the skin mid-plane. With $\mathbf{A}_0$, $\mathbf{B}_0$, $\mathbf{D}_0$ of the combined stack about its own mid-plane and e the distance from the skin mid-plane to that of the stack, the parallel-axis transformation gives

$$\mathbf{A}_e = \mathbf{A}_0, \qquad \mathbf{B}_e = \mathbf{B}_0 + e\,\mathbf{A}_0, \qquad \mathbf{D}_e = \mathbf{D}_0 + 2e\,\mathbf{B}_0 + e^2\mathbf{A}_0. \tag{3}$$

3.2. Cross-section discretisation and stringer–skin tie

The mid-line of the cell is divided into walls (skin, feet, webs, crowns) and the walls into segments of length at most $b_s/25$, with at least two segments per wall. The end points of the walls are the *natural nodes* and the remaining segment end points are *intermediate nodes* (Figure 1b), as in classical GBT [24,35]. For $b_s = 150$ mm the cell has 92 nodes, 19 walls and 93 segments. On a segment of length l between nodes a and b, with $\eta = (s - s_a)/l \in [0, 1]$, the profiles $\bar{u}$ and $\bar{v}$ are linear, and $\bar{w}$ is the cubic Hermite interpolation of the nodal normal displacements and the nodal rotations $\theta = \bar{w}_{,s}$:

$$\bar{w}(\eta) = H_1(\eta)\,\bar{w}_a + l\,H_2(\eta)\,\theta_a + H_3(\eta)\,\bar{w}_b + l\,H_4(\eta)\,\theta_b, \qquad \begin{aligned}H_1 &= 1 - 3\eta^2 + 2\eta^3, & H_2 &= \eta - 2\eta^2 + \eta^3,\\ H_3 &= 3\eta^2 - 2\eta^3, & H_4 &= -\eta^2 + \eta^3.\end{aligned} \tag{4}$$

The nodal rotation is shared by all walls that meet at a node (rigid fold lines). The in-plane displacement of node i is the vector $\mathbf{U}_i = (U_{X,i}, U_{Y,i})$; on a segment with unit tangent **t** and unit normal **n**, $\bar{v} = \mathbf{t}\cdot\mathbf{U}$ and $\bar{w} = \mathbf{n}\cdot\mathbf{U}$ at its end nodes. The space spanned by Eq. (4) is the same as that of the classical description by nodal transverse moments (Schardt's plate-strip flexibility formulation used in [43]); the nodal Hermite form is used because it makes the periodic constraints of Section 3.4 linear rows on nodal quantities. All cross-section integrals use four Gauss points per segment.

In the shell model the stringer mid-surface is tied to the skin with the offset c. In GBT, every web root is a node at $Y = c$, joined to the skin node below it by a *link wall* of height c. The link wall must transmit the in-plane displacements and the rotation like a line of rigid offset links, but it must not add axial or twisting stiffness. Its laminate is therefore built from the skin stiffnesses $\mathbf{A}_0$, $\mathbf{D}_0$, $\mathbf{A}_{s0}$ as

$$
\begin{aligned}
&A^L_{11} = 10^{-3} A_{0,11}, \quad A^L_{22} = k_L A_{0,22}, \quad A^L_{66} = k_L A_{0,66}, \\
&D^L_{11} = 10^{-3} D_{0,11}, \quad D^L_{22} = k_L D_{0,22}, \quad D^L_{66} = 10^{-3} D_{0,66}, \quad \mathbf{A}^L_s = k_L \mathbf{A}_{s0}
\end{aligned}
\tag{5}
$$

with $k_L = 10^3$ and all other entries zero; link walls carry no load. The result does not depend on k_L: for the isotropic SS panel with $b_s = 150$ mm and $\beta = 1.5$, $N_{cr} = 122.656$, 122.663 and 122.660 N/mm for $k_L = 10^2$, 10^3 and 10^4. A link wall with a stiff twisting term D^L_{66} locks the root rotation and over-predicts N_{cr} by 23 %; omitting it is the consistent choice for a row of rigid links.

3.3. Kinematics and deformation mode families

3.3.1. Displacement field

The displacement field of shear-deformable GBT [31,36] is extended here by the transverse shear rotations of the Reissner–Mindlin walls. The mid-surface displacements and the transverse shear rotations are

$$
\begin{aligned}
u(x,s) &= \sum_{k\in\mathcal{C}} \bar{u}_k(s)\,\varphi_{k,x}(x) + \varphi_a(x) + \sum_{j\in\mathcal{S}} \bar{u}_j(s)\,\varphi_j(x), \\
v(x,s) &= \sum_{k\in\mathcal{C}} \bar{v}_k(s)\,\varphi_k(x), \qquad w(x,s) = \sum_{k\in\mathcal{C}} \bar{w}_k(s)\,\varphi_k(x), \\
\beta_x(x,s) &= \sum_{m\in\mathcal{M}_x} \bar{c}_m(s)\,\chi_m(x), \qquad \beta_s(x,s) = \sum_{m\in\mathcal{M}_s} \bar{b}_m(s)\,\vartheta_m(x),
\end{aligned}
\tag{6}
$$

where $\mathcal{C}$ is the set of conventional modes, whose warping $\bar{u}_k$ follows from their in-plane deformation, φ_a the amplitude of the uniform axial mode, $\mathcal{S}$ the set of shear (warping) modes with independent amplitudes φ_j, and $\mathcal{M}_x$, $\mathcal{M}_s$ the sets of transverse shear (Mindlin) modes. The rotations of the wall normal are $\phi_x = -w_{,x} + \beta_x$ and $\phi_s = -w_{,s} + \beta_s$, so that the in-plane displacements at a distance z from the mid-surface are $u + z\phi_x$ and $v + z\phi_s$. For $\beta_x = \beta_s = 0$ the Kirchhoff kinematics of classical GBT is recovered.

3.3.2. Strain–displacement relations

The linear strains of a flat wall are

$$
\begin{aligned}
&\varepsilon_{xx} = u_{,x}, && \varepsilon_{ss} = v_{,s}, && \gamma_{xs} = u_{,s} + v_{,x}, \\
&\kappa_{xx} = \phi_{x,x} = -w_{,xx} + \beta_{x,x}, && \kappa_{ss} = \phi_{s,s} = -w_{,ss} + \beta_{s,s}, && \kappa_{xs} = \phi_{x,s} + \phi_{s,x} = -2w_{,xs} + \beta_{x,s} + \beta_{s,x}, \\
&\gamma_{xz} = w_{,x} + \phi_x = \beta_x, && \gamma_{sz} = w_{,s} + \phi_s = \beta_s.
\end{aligned}
\tag{7}
$$

Collecting all amplitude functions in $\mathbf{q}(x) = \left[\varphi_{\mathcal{C}};\ \varphi_a;\ \varphi_{\mathcal{S}};\ \chi;\ \vartheta\right]$ (n modes) and inserting Eq. (6) into Eq. (7) gives

$$
\widehat{\varepsilon}(x,s) = \sum_{d=0}^{2} \mathbf{E}^{(d)}(s)\,\mathbf{q}^{(d)}(x), \qquad \mathbf{q}^{(d)} = \frac{d^d \mathbf{q}}{dx^d},
\tag{8}
$$

where the columns $\mathbf{E}^{(d)}_k(s)$ are the generalised strains produced by a unit d-th derivative of the amplitude of mode k. Their non-zero entries are listed in Table 2. The conventional rows contain the classical GBT strains; the entries $\varepsilon_{ss} = \bar{v}_{k,s}$ belong to the transverse-extension modes, and the entries of the shear and Mindlin modes are those of shear-deformable GBT.

Table 2. Non-zero generalised strains $\mathbf{E}^{(d)}_k$ of one mode for a unit d-th derivative of its amplitude.

Mode (amplitude)	$d = 0$	$d = 1$	$d = 2$
conventional (φ_k)	$\varepsilon_{ss} = \bar{v}_{k,s}$, $\kappa_{ss} = -\bar{w}_{k,ss}$	$\gamma_{xs} = \bar{u}_{k,s} + \bar{v}_k$, $\kappa_{xs} = -2\bar{w}_{k,s}$	$\varepsilon_{xx} = \bar{u}_k$, $\kappa_{xx} = -\bar{w}_k$
axial (φ_a)	–	$\varepsilon_{xx} = 1$	–

Mode (amplitude)	$d = 0$	$d = 1$	$d = 2$
shear (φ_j)	$\gamma_{xs} = \bar{u}_{j,s}$	$\varepsilon_{xx} = \bar{u}_j$	–
Mindlin, β_x (χ_m)	$\kappa_{xs} = \bar{c}_{m,s}$, $\gamma_{xz} = \bar{c}_m$	$\kappa_{xx} = \bar{c}_m$	–
Mindlin, β_s (ϑ_m)	$\kappa_{ss} = \bar{b}_{m,s}$, $\gamma_{sz} = \bar{b}_m$	$\kappa_{xs} = \bar{b}_m$	–

3.3.3. Deformation mode families

Let N be the number of nodes. The in-plane nodal displacements $\mathbf{U} = [\mathbf{U}_1; \dots; \mathbf{U}_N] \in \mathbb{R}^{2N}$ and the nodal rotations $\Theta \in \mathbb{R}^N$ describe the in-plane deformation of the cross-section. Let $\mathbf{E}_\ell$ be the matrix that gives the elongation of every segment, $(\mathbf{E}_\ell \mathbf{U})_r = \mathbf{t}_r \cdot (\mathbf{U}_b - \mathbf{U}_a)$. Following the organisation of the mode families of [35,36], the conventional modes are defined by

- **G (global):** the rigid in-plane translations and rotation of the cross-section, as far as the side conditions admit them;
- **D (distortional):** in-plane nodal displacement fields without segment elongation, $\mathbf{E}_\ell \mathbf{U} = \mathbf{0}$ (Vlasov's null transverse membrane strain), orthogonal to G and L;
- **L (local):** a unit normal displacement of one node that is not a wall junction, and one pure nodal rotation per node (the in-plane fields of G, D and T carry no nodal rotations, so the rotations are independent coordinates);
- **T (transverse extension):** segment elongation patterns, the columns of $\mathbf{E}_\ell^{\mathsf{T}}$;

and the non-conventional modes by

- **A (axial):** the uniform axial displacement $u = \varphi_a$;
- **S (shear):** the nodal hat functions $h_i(s)$ of the warping $\bar{u}_j$ (one is dropped, since their sum is the axial mode);
- **M (Mindlin):** for β_x, a nodal vector Ω_i in the cross-section plane, with $\bar{c}_m = \mathbf{t} \cdot \Omega$ on every segment (one vector at a node inside a straight wall, two at a fold line); for β_s, one nodal scalar per node.

Since $\mathbb{R}^{2N} = \ker \mathbf{E}_\ell \oplus \operatorname{range} \mathbf{E}_\ell^{\mathsf{T}}$, the families G, D, L and T together span all in-plane nodal displacements, and the rotation columns make the nodal rotations independent. The conventional modes are therefore a complete nodal description of the in-plane deformation, organised in the GBT families; the mode counts are given in Table 3.

Warping of the global and distortional modes. In classical GBT, the warping $\bar{u}_k$ of the G and D modes follows from Vlasov's condition of null membrane shear strain [24,52], $\gamma_{xs} = \bar{u}_{k,s} + \bar{v}_k = 0$ on every wall. The omega stringer and the skin form closed cells, where this condition cannot be satisfied exactly. The warping is therefore determined as the stiffness-weighted least-squares solution

$$\min_{\bar{\mathbf{u}}} \sum_r \hat{A}_{66,r}\, l_r \left(\frac{\bar{u}_{b(r)} - \bar{u}_{a(r)}}{l_r} + \bar{v}_r\big|_{\eta=1/2} \right)^2 \quad \text{subject to} \quad \mathbf{c}_{ax}^{\mathsf{T}} \bar{\mathbf{u}} = 0, \tag{9}$$

with the sum over all segments r of length l_r between the nodes $a(r)$ and $b(r)$, $\hat{A}_{66} = A_{66} - A_{26}^2/A_{22}$ the membrane shear stiffness after condensing $N_{ss} = 0$, and $c_{ax,i} = \sum_{r \ni i} \frac{1}{2} \hat{A}_{11,r}\, l_r$ with $\hat{A}_{11} = A_{11} - A_{12}^2/A_{22}$, so that the stiffness-weighted mean warping vanishes (no axial content). For an open section Eq. (9) reproduces the classical warping exactly (zero residual); in the closed cells the residual membrane shear is carried by the shear modes S.

3.4. Periodic boundary conditions in GBT

3.4.1. The periodic condition

The cell is one period of an infinitely wide panel. The displacement field at the side edge $X = -W$ (node s_L) must therefore equal that at $X = 0$ (node s_R), for all components and at every x:

$$\mathcal{U}_\alpha(X = -W,\, x) = \mathcal{U}_\alpha(X = 0,\, x), \qquad \alpha = 1, \dots, 6, \qquad 0 \le x \le L, \tag{10}$$

where $\mathcal{U}_\alpha$ are the six degrees of freedom of a shell node (three translations, three rotations). This is the condition that the Abaqus model imposes with one constraint equation per pair of side nodes and degree of

freedom [5–7]. On the skin side edges, whose normal is Y, the shell degrees of freedom correspond to the GBT quantities in Table 4. Since all of them are linear in the amplitude functions, Eq. (10) holds for every x and every set of amplitude functions if and only if *every mode* satisfies it. The conditions are therefore built into the modes, family by family.

Table 3. Number of deformation modes per family for the two side conditions (Cross-ply lay-up; the counts do not depend on the lay-up).

b_s [mm]	side edges	G	D	L	T	A	S	M	total
150	symmetry	2	10	166	94	1	91	205	569
150	periodic	2	13	166	92	1	90	205	569
285.5	symmetry	2	10	138	80	1	77	174	482
285.5	periodic	2	13	138	78	1	76	174	482

Table 4. Degrees of freedom of a skin side node and the corresponding GBT conditions (Eqs. (11)–(15)).

Shell DOF	Meaning at a skin side node	Periodic condition in GBT
U_1	in-plane displacement U_X	$U_{X,s_L} = U_{X,s_R}$ (row of $\mathbf{R}$)
U_2	deflection $U_Y = w$	$U_{Y,s_L} = U_{Y,s_R}$ (row of $\mathbf{R}$)
U_3	axial displacement u	warping of G/D modes $\bar{u}_{s_L} = \bar{u}_{s_R}$; shear hats of s_L, s_R merged; axial mode uniform
UR_1	rotation $-w_{,x} + \beta_x$	$w_{,x}$ periodic because U_Y is; $\Omega_{s_L} = \Omega_{s_R}$
UR_3	rotation $w_{,s} + \beta_s = \theta + \beta_s$	$\theta_{s_L} = \theta_{s_R}$ (row of $\mathbf{R}$); $\bar{b}_{s_L} = \bar{b}_{s_R}$
UR_2	drilling rotation	not a degree of freedom of the plate theory

3.4.2. Conventional modes

For the conventional modes the conditions $U_{X,s_L} = U_{X,s_R}$, $U_{Y,s_L} = U_{Y,s_R}$ and $\theta_{s_L} = \theta_{s_R}$ are the rows of a constraint matrix acting on the stacked nodal vector:

$$\mathbf{R}\begin{bmatrix}\mathbf{U}\\ \Theta\end{bmatrix} = \mathbf{0}, \qquad \mathbf{R} = \begin{bmatrix}\mathbf{e}_{X,s_L}^{\mathsf{T}} - \mathbf{e}_{X,s_R}^{\mathsf{T}}\\ \mathbf{e}_{Y,s_L}^{\mathsf{T}} - \mathbf{e}_{Y,s_R}^{\mathsf{T}}\\ \mathbf{e}_{\theta,s_L}^{\mathsf{T}} - \mathbf{e}_{\theta,s_R}^{\mathsf{T}}\end{bmatrix} \in \mathbb{R}^{3\times 3N}, \tag{11}$$

where $\mathbf{e}_{X,i}$, $\mathbf{e}_{Y,i}$ and $\mathbf{e}_{\theta,i}$ are the unit vectors that extract $U_{X,i}$, $U_{Y,i}$ and θ_i. Let $\mathbf{B}_g \in \mathbb{R}^{3N\times n_g}$ hold the nodal fields $[\mathbf{U};\Theta]$ of family $g \in \{G, D, L, T\}$ of the unconstrained cell. The constraint is imposed family by family, so that the mechanical meaning of the families is kept:

1. the columns of $\mathbf{B}_g$ that satisfy $\mathbf{Rb} = \mathbf{0}$ are kept unchanged;
2. the remaining columns are replaced by a basis of the null space of the family, $\mathbf{B}_g \leftarrow \mathbf{B}_g\mathbf{Z}_g$ with $\mathbf{Z}_g = \ker(\mathbf{RB}_g)$;
3. the vectors of $\ker\left(\mathbf{R}\left[\mathbf{B}_G\ \mathbf{B}_D\ \mathbf{B}_L\ \mathbf{B}_T\right]\right)$ that are not yet spanned (combinations of several families) are added, to the family T if they elongate a segment ($\mathbf{E}_\ell\mathbf{U} \neq \mathbf{0}$) and to the family D otherwise;
4. each column is scaled to a unit maximum nodal displacement or rotation.

The resulting modal matrix $\mathbf{B} = [\mathbf{B}_G\ \mathbf{B}_D\ \mathbf{B}_L\ \mathbf{B}_T]$ satisfies $\mathbf{RB} = \mathbf{0}$ and spans the whole periodic subspace $\ker\mathbf{R}$. Consequently, for arbitrary amplitude functions,

$$\mathbf{R}\sum_{k\in\mathcal{C}}\begin{bmatrix}\mathbf{U}_k\\ \Theta_k\end{bmatrix}\varphi_k^{(d)}(x) = \mathbf{0} \qquad \text{for all } x \text{ and } d = 0, 1, 2, \tag{12}$$

and the same holds for any subset of the modes. The rigid rotation of the cross-section about the x-axis moves the two side nodes in opposite directions ($U_{Y,s_L} - U_{Y,s_R} = W\omega$ for a rotation ω) and drops out:

a torsional rigid-body motion of one cell is not compatible with an infinitely wide panel. The two rigid translations remain (the two G modes in Table 3). Compared with the symmetry side edges, where the rows are $U_X = \theta = 0$ at both side nodes, the periodic cell has more distortional modes (13 instead of 10 in Table 3): the side nodes may move laterally and rotate, provided that they do so identically.

3.4.3. Warping of the global and distortional modes

For the warping of the G and D modes, the periodic condition $\bar{u}_{s_L} = \bar{u}_{s_R}$ is added to the constraint of Eq. (9). With the segment difference matrix $\mathbf{H}$, $(\mathbf{H}\bar{\mathbf{u}})_r = \bar{u}_{b(r)} - \bar{u}_{a(r)}$, the weight matrix $\mathbf{W} = \mathrm{diag}(\hat{A}_{66,r}/l_r)$, the vector $\mathbf{g}$ of the segment mid-point values $l_r\bar{v}_r(1/2)$, and the constraint matrix $\mathbf{C}_u = \left[\mathbf{c}_{ax}^{\mathsf{T}};\ \mathbf{e}_{s_L}^{\mathsf{T}} - \mathbf{e}_{s_R}^{\mathsf{T}}\right]$, the warping of each mode follows from one Karush–Kuhn–Tucker system:

$$\begin{bmatrix} \mathbf{H}^{\mathsf{T}}\mathbf{W}\mathbf{H} & \mathbf{C}_u^{\mathsf{T}} \\ \mathbf{C}_u & \mathbf{0} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{u}} \\ \mu \end{bmatrix} = \begin{bmatrix} -\mathbf{H}^{\mathsf{T}}\mathbf{W}\mathbf{g} \\ \mathbf{0} \end{bmatrix}. \tag{13}$$

3.4.4. Shear, Mindlin and axial modes

The shear modes are the nodal hat functions $h_i(s)$ of the warping. For periodic side edges the hat functions of the two side nodes are merged into one mode with a common warping value,

$$h_{s_R} \leftarrow h_{s_R} + h_{s_L}, \qquad h_{s_L} \text{ removed.} \tag{14}$$

The Mindlin modes are nodal as well; the nodal functions of s_L are added to those of s_R and removed:

$$\Omega_{s_R} \leftarrow \Omega_{s_R} + \Omega_{s_L}, \qquad \bar{b}_{s_R} \leftarrow \bar{b}_{s_R} + \bar{b}_{s_L}, \qquad \Omega_{s_L},\ \bar{b}_{s_L} \text{ removed.} \tag{15}$$

The axial mode is uniform and periodic by construction. With symmetry side edges instead, the shear and Mindlin modes of the side nodes are kept separately and the β_s modes of the side nodes are removed ($UR_3 = 0$).

3.4.5. Remarks

Exactness. Equations (11)–(15) make every mode satisfy Eq. (10). Any linear combination with arbitrary amplitude functions, and in particular any truncated subset of the modes, therefore satisfies the periodic condition exactly. No penalty terms or Lagrange multipliers appear in the member analysis. For the cross-ply cell with $b_s = 150$ mm the largest difference between the two side nodes over all conventional modes is 7.9×10^{-15} for the in-plane displacements, 1.1×10^{-16} for the rotation and 2.3×10^{-11} for the warping (unit-normalised modes), i.e. zero to round-off.

Why the condition cannot be added to classical modes afterwards. Classical GBT generates the conventional modes from a few natural nodes of an open contour. The periodic condition couples the two ends of the contour, so that the cell behaves like a closed loop with respect to its in-plane kinematics: the rigid rotation is no longer admissible, the warping must take the same value on both sides of the cut, and the classical hierarchy has to be recomputed. Imposed afterwards on a truncated set of classical modes, the three conditions of Eq. (11) remove three degrees of freedom from a small space and leave too few admissible modes. Here the modes are generated from a complete nodal basis and the constraint is applied before any truncation.

Relation to Bloch–Floquet theory. For an infinite panel with stringer pitch b_s, the buckling modes are Bloch waves, $\mathbf{u}(X + b_s, x) = e^{\mathrm{i}\kappa b_s}\mathbf{u}(X, x)$ [12]. A periodic cell of width $2b_s$ admits exactly the two real cases $\kappa b_s = 0$ (the mode repeats in every bay) and $\kappa b_s = \pi$ (the mode alternates in sign from bay to bay). These are the two patterns that a two-bay cell can represent, and the same restriction applies to the Abaqus model with periodic side edges used for validation.

Hierarchical modes. For visualisation and for reduced bases, the conventional modes can be ordered as in classical GBT by the generalised eigenvalue problem $\bar{\mathbf{B}}\,\psi = \Lambda\,(k_r^2\bar{\mathbf{D}}_1 + k_r^4\bar{\mathbf{C}})\,\psi$ at a reference wavenumber k_r, with the modal matrices of Section 3.5. Small Λ means a preference for long waves (global, then distortional, then local). The first eight hierarchical modes of the periodic cell are shown in Figure 2. The two side nodes move identically in every mode, including the lateral rigid translation (mode 2), while the rigid rotation is absent.

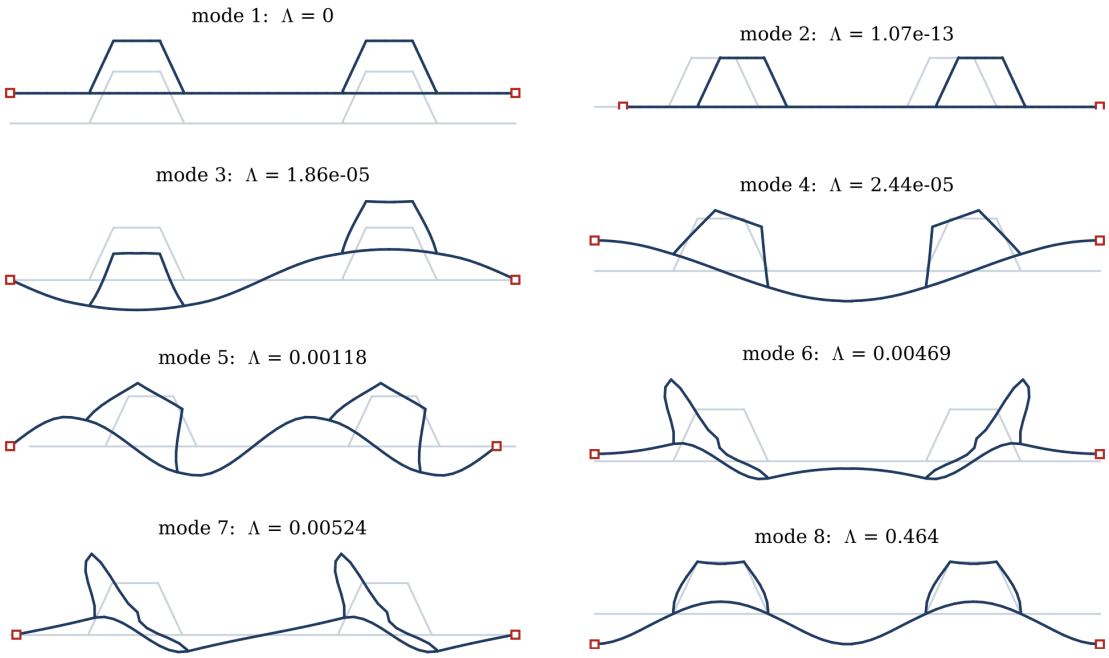


Figure 2. First eight hierarchical conventional modes of the periodic cell ($b_s = 150$ mm, Cross-ply, $k_r = 2\pi/b_s$; in-plane deformation, grey: undeformed). The side nodes s_L and s_R (squares) move identically in every mode.

3.5. Strain energy and GBT modal matrices

With Eq. (8), the strain energy of the cell becomes

$$U = \frac{1}{2}\int_0^L \int_\Gamma \hat{\varepsilon}^{\mathsf{T}}\mathbf{C}\,\hat{\varepsilon}\,ds\,dx = \frac{1}{2}\int_0^L \sum_{d=0}^{2}\sum_{e=0}^{2} \mathbf{q}^{(d)\mathsf{T}}\mathbf{A}^{(de)}\,\mathbf{q}^{(e)}\,dx,$$
$$\mathbf{A}^{(de)} = \int_\Gamma \mathbf{E}^{(d)\mathsf{T}}\,\mathbf{C}\,\mathbf{E}^{(e)}\,ds, \tag{16}$$

where Γ is the mid-line of the cross-section and $\mathbf{A}^{(ed)} = \mathbf{A}^{(de)\mathsf{T}}$. In matrix form, with $\Phi = \left[\mathbf{q};\ \mathbf{q}_{,x};\ \mathbf{q}_{,xx}\right]$,

$$U = \frac{1}{2}\int_0^L \Phi^{\mathsf{T}}\,\Xi\,\Phi\,dx,$$
$$\Xi = \begin{bmatrix} \bar{\mathbf{B}} & \bar{\mathbf{F}}_1 & \bar{\mathbf{D}}_2 \\ \bar{\mathbf{F}}_1^{\mathsf{T}} & \bar{\mathbf{D}}_1 & \bar{\mathbf{H}}_1 \\ \bar{\mathbf{D}}_2^{\mathsf{T}} & \bar{\mathbf{H}}_1^{\mathsf{T}} & \bar{\mathbf{C}} \end{bmatrix} = \begin{bmatrix} \mathbf{A}^{(00)} & \mathbf{A}^{(01)} & \mathbf{A}^{(02)} \\ \mathbf{A}^{(10)} & \mathbf{A}^{(11)} & \mathbf{A}^{(12)} \\ \mathbf{A}^{(20)} & \mathbf{A}^{(21)} & \mathbf{A}^{(22)} \end{bmatrix}, \tag{17}$$

which is the GBT matrix Ξ of the GBT–Ritz approach [43,46] written for the full set of modes. For the conventional modes $k, l \in \mathcal{C}$, Table 2 and Eq. (2) give the classical GBT modal matrices of a laminated wall with coupling:

$$\bar{B}_{kl} = \int_\Gamma \left[A_{22}\,\bar{v}_{k,s}\bar{v}_{l,s} - B_{22}\left(\bar{v}_{k,s}\bar{w}_{l,ss} + \bar{w}_{k,ss}\bar{v}_{l,s}\right) + D_{22}\,\bar{w}_{k,ss}\bar{w}_{l,ss}\right] ds, \tag{18}$$

$$\bar{C}_{kl} = \int_\Gamma \left[A_{11}\,\bar{u}_k\bar{u}_l - B_{11}\left(\bar{u}_k\bar{w}_l + \bar{w}_k\bar{u}_l\right) + D_{11}\,\bar{w}_k\bar{w}_l\right] ds, \tag{19}$$

$$\bar{D}_{1,kl} = \int_\Gamma \left[A_{66}\,\gamma_k\gamma_l - 2B_{66}\left(\gamma_k\bar{w}_{l,s} + \bar{w}_{k,s}\gamma_l\right) + 4D_{66}\,\bar{w}_{k,s}\bar{w}_{l,s}\right] ds,$$
$$\gamma_k = \bar{u}_{k,s} + \bar{v}_k, \tag{20}$$

$$\bar{D}_{2,kl} = \int_\Gamma \Big[A_{12}\,\bar{v}_{k,s}\bar{u}_l - B_{12}\left(\bar{v}_{k,s}\bar{w}_l + \bar{w}_{k,ss}\bar{u}_l\right) + D_{12}\,\bar{w}_{k,ss}\bar{w}_l \Big] ds, \tag{21}$$

$$\bar{F}_{1,kl} = \int_\Gamma \Big[A_{26}\,\bar{v}_{k,s}\gamma_l - B_{26}\left(2\bar{v}_{k,s}\bar{w}_{l,s} + \bar{w}_{k,ss}\gamma_l\right) + 2D_{26}\,\bar{w}_{k,ss}\bar{w}_{l,s} \Big] ds, \tag{22}$$

$$\bar{H}_{1,kl} = \int_\Gamma \Big[A_{16}\,\gamma_k\bar{u}_l - B_{16}\left(\gamma_k\bar{w}_l + 2\bar{w}_{k,s}\bar{u}_l\right) + 2D_{16}\,\bar{w}_{k,s}\bar{w}_l \Big] ds. \tag{23}$$

The transverse-extension stiffness enters through the A_{22} term of $\bar{\mathbf{B}}$, the bending–twisting coupling through $\bar{\mathbf{F}}_1$ and $\bar{\mathbf{H}}_1$, and the membrane–bending coupling through all B_{ij} terms. For the shear modes $j, j' \in \mathcal{S}$, Table 2 gives

$$A^{(00)}_{jj'} = \int_\Gamma A_{66}\,\bar{u}_{j,s}\bar{u}_{j',s}\,ds, \qquad A^{(01)}_{jj'} = \int_\Gamma A_{16}\,\bar{u}_{j,s}\bar{u}_{j'}\,ds, \qquad A^{(11)}_{jj'} = \int_\Gamma A_{11}\,\bar{u}_j\bar{u}_{j'}\,ds, \tag{24}$$

which correspond to the matrices $\breve{\mathbf{D}}_1$, $\breve{\mathbf{H}}_1$ and $\breve{\mathbf{C}}$ of shear-deformable GBT [31,43], while the conventional–shear coupling blocks are the matrices $\tilde{\mathbf{F}}$, $\tilde{\mathbf{D}}$, $\tilde{\mathbf{H}}$ and $\tilde{\mathbf{C}}$ of the same references, evaluated with the strains of Table 2 (e.g. $A^{(12)}_{jk} = \int_\Gamma [A_{11}\bar{u}_j\bar{u}_k - B_{11}\bar{u}_j\bar{w}_k]ds$ for the extension of a shear mode against the warping and bending of a conventional mode). The Mindlin modes add the transverse shear energy $\int_\Gamma [\bar{c}_m A_{s,11}\bar{c}_{m'} + \bar{b}_m A_{s,22}\bar{b}_{m'} + \dots]ds$ and their bending and twisting contributions through $\mathbf{D}$ and $\mathbf{B}$. In the implementation, all blocks are evaluated by the same Gauss quadrature over the segments from the operators $\mathbf{E}^{(d)}$ of Table 2, so that no coupling term is omitted.

3.6. Amplitude functions and end conditions

3.6.1. Ritz approximation along the cell

As in the GBT–Ritz approach [43,46], the amplitude functions are approximated by admissible functions with unknown coefficients, $\Phi = \Psi\,\mathbf{a}$. Instead of global trigonometric series, cubic B-splines on an open, uniform knot vector are used,

$$q_k(x) = \sum_{b=1}^{n_b} N_b(x)\,a_{kb}, \tag{25}$$

with spans of length $h_x = \min(15\text{ mm},\ \max(2\text{ mm},\ W/40),\ L/10)$, i.e. $h_x = b_s/20$ for all cases studied. Local support is essential here, because the pre-buckling state of the cell varies strongly near the loaded edges (Section 3.7), which a single trigonometric series cannot represent efficiently, and because the B-splines represent simply supported and clamped edges alike. Inserting Eq. (25) into Eq. (16) gives the stiffness matrix as a sum of Kronecker products,

$$\mathbf{K} = \sum_{d=0}^{2}\sum_{e=0}^{2} \mathbf{A}^{(de)} \otimes \mathbf{X}^{(de)}, \qquad X^{(de)}_{ab} = \int_0^L \frac{d^d N_a}{dx^d}\,\frac{d^e N_b}{dx^e}\,dx, \tag{26}$$

with four Gauss points per span. For $b_s = 150$ mm and $L = 225$ mm the model has $569 \times 33 = 18\,777$ coefficients. With open knots, $q_k(0) = a_{k1}$ and $q_k(L) = a_{kn_b}$, and $q_{k,x}(0) = 0$ is equivalent to $a_{k2} = a_{k1}$, so that the essential end conditions act directly on the end coefficients.

3.6.2. End conditions

The loaded edges are supported as in the Abaqus models:

- **SS** (out-of-plane support of every end edge, $U_2 = 0$): $U_Y = 0$ at every node at $x = 0$ and $x = L$, i.e. $\sum_{k\in\mathcal{C}} U_{Y,ik}\,\varphi_k(0) = \sum_{k\in\mathcal{C}} U_{Y,ik}\,\varphi_k(L) = 0$ for every node i. These linear conditions on the end coefficients are imposed by a null-space reduction; the in-plane displacement U_X, the warping and the rotations remain free. Rigid-body motions are removed by $\varphi_a(0) = 0$ and, for periodic side edges, $U_X(s_R, 0) = 0$.
- **CC** (clamped loaded edges): $\varphi_k = \varphi_{k,x} = 0$ at both ends for every conventional mode (in-plane displacements and the rotations $w_{,x}$ and θ vanish), $\chi_m = \vartheta_m = 0$ at both ends, and $\varphi_a(0) = \varphi_j(0) = 0$, so that $u = 0$ on the whole edge $x = 0$; the edge $x = L$ is free to warp under the uniform load.

In classical GBT, simply supported ends are represented by $\varphi_k = \sin(m\pi x/L)$, which suppresses all in-plane displacements of the end cross-sections while leaving their warping free (rigid end diaphragms). The present SS condition is weaker: like the Abaqus model it only suppresses the out-of-plane displacement U_Y of the end edges. The classical end condition is recovered by prescribing $\varphi_k(0) = \varphi_k(L) = 0$ for all conventional modes.

3.7. Loads and pre-buckling state

The external work of the end loads, a line load $q(s)$ per unit length acting at a distance $e(s)$ from the wall reference surface, is

$$W_{ext} = \int_\Gamma q(s)\,\big[u_q(s,0) - u_q(s,L)\big]\,ds, \qquad u_q = u - e(s)\,w_{,x}, \tag{27}$$

where u_q is the axial displacement of the fibre at which the load acts ($e = c/2$ on a foot, $e = 0$ elsewhere). Inserting Eqs. (6) and (25) gives the load vector

$$\begin{aligned} f_{\varphi_k,b} &= \left(\int_\Gamma q\,\bar{u}_k\,ds - \int_\Gamma q\,e\,\bar{w}_k\,ds\right)\big[N_{b,x}(0) - N_{b,x}(L)\big]\,, \\ f_{\varphi_j,b} &= \int_\Gamma q\,\bar{u}_j\,ds\,[N_b(0) - N_b(L)]\,, \qquad f_{\varphi_a,b} = \int_\Gamma q\,ds\,[N_b(0) - N_b(L)] \end{aligned} \tag{28}$$

The pre-buckling state is not assumed: it is the first-order solution $\mathbf{K}\mathbf{a}^0 = \mathbf{f}$ under the end and side conditions above. The membrane resultants of this state, compression positive, are

$$[P_{xx},\ P_{ss},\ P_{xs}]^{\mathsf{T}} = -\left(\mathbf{A}\,\varepsilon_m^0 + \mathbf{B}\,\kappa^0\right), \tag{29}$$

with the membrane strains ε_m^0 and curvatures κ^0 of $\mathbf{a}^0$. They vary along s and x: the load is introduced at the ends and shared between skin and stringer by shear lag, the eccentric foot load bends the cell near the ends, and the restrained Poisson expansion produces a transverse resultant P_{ss}. Periodic and symmetry side edges both prevent a net lateral expansion of the cell and therefore both produce P_{ss}; they differ in the shear and rotation of the side edges.

3.8. Geometric stiffness and eigenvalue problem

3.8.1. Geometric stiffness

The second-order membrane energy of the pre-buckling resultants is

$$\begin{aligned} U_G &= \frac{1}{2}\int_0^L \int_\Gamma \sum_{c\in\{u,v,w\}} \Big[P_{xx}\,g_{c,x}^2 + P_{ss}\,g_{c,s}^2 + 2P_{xs}\,g_{c,x}\,g_{c,s}\Big]\,ds\,dx, \\ g_{c,\alpha}(x,s) &= \sum_k \sum_d G_{k,c\alpha}^{(d)}(s)\,q_k^{(d)}(x), \end{aligned} \tag{30}$$

where $g_{c,\alpha}$ is the derivative of the displacement component c with respect to $\alpha \in \{x, s\}$. For a conventional mode, $g_{u,x} = \bar{u}_k\varphi_{k,xx}$, $g_{u,s} = \bar{u}_{k,s}\varphi_{k,x}$, $g_{v,x} = \bar{v}_k\varphi_{k,x}$, $g_{v,s} = \bar{v}_{k,s}\varphi_k$, $g_{w,x} = \bar{w}_k\varphi_{k,x}$ and $g_{w,s} = \bar{w}_{k,s}\varphi_k$. All three displacement components and all three resultants are kept. With Eq. (30),

$$U_G = \frac{1}{2}\int_0^L \sum_{d,e} \mathbf{q}^{(d)\mathsf{T}}\, \mathcal{X}^{(de)}(x)\, \mathbf{q}^{(e)}\, dx,$$
$$\mathcal{X}^{(de)}_{kl}(x) = \sum_t \int_\Gamma P_t(s,x) \sum_c G^{(d)}_{k,c\alpha_t}(s)\, G^{(e)}_{l,c\beta_t}(s)\, ds, \tag{31}$$

with $(P_t, \alpha_t, \beta_t) \in \{(P_{xx}, x, x), (P_{ss}, s, s), (P_{xs}, x, s), (P_{xs}, s, x)\}$. Since the resultants depend on x, the geometric matrices $\mathcal{X}^{(de)}(x)$ are evaluated at every Gauss point along the length, and the geometric stiffness matrix is

$$\mathbf{K}_G = \sum_{d,e} \int_0^L \left(\mathbf{N}^{(d)}\mathbf{N}^{(e)\mathsf{T}}\right) \otimes \mathcal{X}^{(de)}(x)\, dx, \tag{32}$$

with $\mathbf{N}^{(d)}$ the vector of the d-th derivatives of the B-splines. The Mindlin modes do not move the mid-surface and do not enter $\mathbf{K}_G$.

3.8.2. Eigenvalue problem and modal participation

The critical load follows from the stationarity of $U - \lambda U_G$:

$$(\mathbf{K} - \lambda\, \mathbf{K}_G)\, \mathbf{a} = \mathbf{0}, \qquad N_{cr} = \lambda_1 \times 1\ \mathrm{N/mm}, \tag{33}$$

on the coefficients that satisfy the end conditions. Since $\mathbf{K}_G$ is indefinite (the pre-buckling state contains membrane shear and some transverse tension), the problem is solved as $\mathbf{K}_G\, \mathbf{a} = \mu\, \mathbf{K}\, \mathbf{a}$ for the largest $\mu = 1/\lambda$, with the implicitly restarted Lanczos method [53] and a sparse direct factorisation of $\mathbf{K}$ (tolerance 10^{-8}).

The participation of a mode family g in the buckling mode is measured as in [46] by the share of the displacement field that the family produces,

$$p_g = \frac{\int_0^L \|\mathbf{F}_g(x)\|\, dx}{\sum_h \int_0^L \|\mathbf{F}_h(x)\|\, dx}, \tag{34}$$

where $\|\mathbf{F}_g(x)\|$ is the root mean square over the cross-section of the displacements produced by family g alone.

3.9. Classical limits

The formulation contains classical GBT as a special case. For the conventional modes alone, uncoupled orthotropic walls ($\mathbf{B} = \mathbf{0}$, $A_{16} = A_{26} = D_{16} = D_{26} = 0$) and a uniform pre-buckling resultant $P_{xx} = N^0$ with $P_{ss} = P_{xs} = 0$, the odd blocks vanish, $\bar{\mathbf{F}}_1 = \mathbf{H}_1 = \mathbf{0}$, and the geometric matrices of Eq. (31) reduce to

$$\mathcal{X}^{(11)} = \mathcal{X}^{VW}, \quad \mathcal{X}^{VW}_{kl} = \int_\Gamma N^0\, (\bar{v}_k \bar{v}_l + \bar{w}_k \bar{w}_l)\, ds,$$
$$\mathcal{X}^{(22)} = \mathcal{X}^{U}, \quad \mathcal{X}^{U}_{kl} = \int_\Gamma N^0\, \bar{u}_k \bar{u}_l\, ds, \tag{35}$$

which are the geometric matrices of the GBT–Ritz approach [43,46]. The Euler–Lagrange equations of $U - \lambda U_G$ with Eqs. (17) and (35) are

$$\bar{\mathbf{C}}\, \varphi_{,xxxx} - \bar{\mathbf{D}}\, \varphi_{,xx} + \bar{\mathbf{B}}\, \varphi + \lambda \left(\mathcal{X}^{VW} \varphi_{,xx} - \mathcal{X}^{U} \varphi_{,xxxx}\right) = \mathbf{0},$$
$$\bar{\mathbf{D}} = \bar{\mathbf{D}}_1 - \bar{\mathbf{D}}_2 - \bar{\mathbf{D}}_2^{\mathsf{T}}, \tag{36}$$

where the integration by parts of the $\bar{\mathbf{D}}_2$ terms gives the classical torsion-related matrix $\bar{\mathbf{D}}$. Without the warping contribution $\mathcal{X}^U$, Eq. (36) is the classical GBT buckling equation $\bar{\mathbf{C}}\varphi'''' - \bar{\mathbf{D}}\varphi'' + \bar{\mathbf{B}}\varphi + \lambda \mathcal{X} \varphi'' = \mathbf{0}$

of Schardt [24] and of Silvestre and Camotim for orthotropic members [30]. For simply supported ends and $\varphi = \hat{\varphi} \sin k_m x$, $k_m = m\pi/L$, it becomes the algebraic eigenproblem

$$\left[\bar{\mathbf{B}} + k_m^2 \bar{\mathbf{D}} + k_m^4 \bar{\mathbf{C}} - \lambda \left(k_m^2 \mathcal{X}^{VW} + k_m^4 \mathcal{X}^{U}\right)\right] \hat{\varphi} = \mathbf{0}, \tag{37}$$

whose lowest eigenvalue over m is the classical GBT signature curve. The periodic condition of Section 3.4 does not alter this structure: it only restricts the admissible modes of the cross-section. Equation (37) also explains the hierarchical ordering used in Figure 2. The present model differs from the classical one only through the terms that the panel requires and that classical GBT omits: laminate coupling ($\bar{\mathbf{F}}_1$, $\bar{\mathbf{H}}_1$, B_{ij}), shear and Mindlin modes, a non-uniform pre-buckling state with P_{ss} and P_{xs}, and end conditions that support the loaded edges without restraining their warping.

3.10. Numerical aspects

Empty modes. The Mindlin vectors Ω_i are an orthonormal basis of the span of the tangents of the segments meeting at node i. At some intermediate nodes of the inclined webs, the two segment tangents are identical up to round-off, and a rank detection without tolerance keeps a spurious second vector normal to the web. For it $\bar{c}_m = \mathbf{t} \cdot \Omega \approx 10^{-15}$ on every segment, so that the mode carries neither strain energy ($A_{kk}^{(de)} \approx 10^{-25}$ against 10^9 for the stiffest mode) nor displacement gradients. There are 11 such modes for $b_s = 150$ mm and 8 for $b_s = 285.5$ mm. They cannot change any eigenvalue, but left free they make $\mathbf{K}$ singular; they are identified by $\sum_{d,e} |A_{kk}^{(de)}| \leq 10^{-12} \max_l \sum_{d,e} |A_{ll}^{(de)}|$ together with $\mathbf{G}_k \equiv \mathbf{0}$ and their coefficients are held. With this treatment the result does not depend on the sparse solver (e.g. 31.2431 N/mm for Cross-ply, SS, periodic, $b_s = 150$ mm, $\beta = 1.5$, with both PARDISO and SuperLU).

Sparse geometric stiffness. At a cross-section Gauss point s_p only the modes with support there are non-zero (about 25 of 569 for $b_s = 150$ mm). Writing Eq. (31) with quadrature, $\mathcal{X}_{kl}^{(de)}(x) = \sum_t \sum_p P_t(s_p, x)\, M_{t,p,(kl)}^{(de)}$, the sparse factors $M_{t,p,(kl)}^{(de)} = w_p \sum_c G_{k,c\alpha_t}^{(d)}(s_p) G_{l,c\beta_t}^{(e)}(s_p)$ depend only on the cross-section and are computed once; for every panel length only $P_t(s_p, x_g)$ changes.

4. Verification and discussion

4.1. Reference models

Abaqus shell models. The reference solutions are linear buckling analyses in Abaqus with S4R and S3 shell elements and a seed of 3.7 mm. The stringer is tied to the skin with *TIE and the true offset of the mid-surfaces, and the loads are *SHELL EDGE LOAD of 1 N/mm (2 N/mm on the feet, Section 2.2) at both loaded edges. SS ends prescribe $U_2 = 0$ on every end edge of skin and stringer; CC ends prescribe $U_1 = U_2 = UR_1 = UR_2 = UR_3 = 0$ at both ends and $U_3 = 0$ at $x = 0$. Symmetry side edges prescribe $U_1 = UR_2 = UR_3 = 0$. For periodic side edges, every skin node of the edge $X = -W$ is paired with the node of $X = 0$ at the same x, and for each pair one constraint equation (*EQUATION) $\mathcal{U}_\alpha(-W, x) - \mathcal{U}_\alpha(0, x) = 0$ is written for each of the six degrees of freedom, as in Eq. (10); at the two end corners the equation for U_2 is omitted because $U_2 = 0$ is already prescribed there. The rigid-body motions left by periodicity are removed by $U_1 = U_3 = 0$ at the corner $X = 0$, $x = 0$. The Abaqus database covers 60 cases with symmetry side edges (five lay-ups, SS and CC, two pitches, three aspect ratios) and the 30 SS cases with periodic side edges. Periodic CC cases have not been computed in Abaqus.

The GBT model imposes the same physical conditions (Table 4 and Section 3.6). Two differences of form do not affect the eigenvalue: Abaqus ties the nodes of its mesh, whereas the GBT modes satisfy the periodic condition at every x; and the drilling rotation UR_2, which Abaqus ties, is not a degree of freedom of the plate theory (in S4R it carries only a small artificial stiffness).

Spline finite strip model. As a second, independent reference, in particular for the periodic CC cases, an in-house spline finite strip model [10,11] is used. It discretises the same geometry, laminates, loads and end conditions with flat laminated Mindlin strips of width $b_s/50$ (twice as fine as the GBT segments), linear interpolation across each strip, a mid-strip transverse shear strain against shear locking, and six degrees of freedom per nodal line in global axes. Along x it uses the same cubic B-splines as GBT. The web roots are joined to the skin by rigid offset links (master–slave constraints), the direct analogue of the Abaqus *TIE, and the side conditions are applied node by node, exactly as in Abaqus: the six degrees of freedom of the two side

nodal lines are merged for every spline coefficient. Pre-buckling state and geometric stiffness are computed as in Sections 3.7 and 3.8. The strip model shares the plate theory with GBT but none of the cross-section discretisation: it has no deformation modes, a finer mesh, rigid links instead of link walls, and boundary conditions on nodal degrees of freedom instead of built into modes. Its agreement with GBT is therefore a genuine check of the mode construction. For $b_s = 150$ mm and $L = 225$ mm it has 37 170 unknowns (177 nodal lines × 6 × 35 splines), against 18 777 for GBT.

4.2. Exactness of the periodic condition

Since the condition is built into the modes, every buckling mode satisfies it identically. For the Cross-ply cell with $b_s = 150$ mm, the largest difference between the side nodes over all conventional modes is 7.9×10^{-15} for the translations, 1.1×10^{-16} for the rotation θ and 2.3×10^{-11} for the warping (unit-normalised modes). In the buckling mode of the Unsym panel with CC ends, $b_s = 150$ mm and $\beta = 1.5$, the largest side-edge differences along the whole length are 1.2×10^{-18} (U_X), 6.9×10^{-18} (U_Y), 1.4×10^{-20} (θ) and 1.1×10^{-15} (warping), with modal amplitudes of order 10^{-5} to 10^{-1}, while the side edges do move ($U_X, \theta \neq 0$). With symmetry side edges the same panel has $U_X = \theta = 0$ at the sides and different deflections of the two side edges.

4.3. Symmetry side edges

Table 5 compares the critical loads for the 60 cases with symmetry side edges; the individual results are listed in Appendix A. GBT agrees with Abaqus within 0.58 % on average and 1.48 % at most, and the spline finite strip model within 0.57 % and 1.30 %. GBT and strip differ by 0.27 % on average. The accuracy is the same for SS and CC ends and does not deteriorate for the coupled lay-ups, which confirms the treatment of the laminate couplings in Eqs. (18)–(23) and the clamped end condition of Section 3.6.

Table 5. Absolute deviation of the critical load in %, 60 cases with symmetry side edges (six cases per row: $b_s = 150, 285.5$ mm, $\beta = 0.75, 1.5, 3$).

Lay-up	Ends	GBT vs Abaqus, mean	max	Strip vs Abaqus, mean	max	GBT vs strip, mean	max
Isotropic	SS	0.56	1.08	0.59	1.10	0.12	0.28
Isotropic	CC	0.58	1.08	0.55	1.05	0.13	0.28
Cross-ply	SS	0.49	0.91	0.41	0.69	0.22	0.35
Cross-ply	CC	0.50	0.92	0.42	0.87	0.14	0.25
Sym1	SS	0.52	0.90	0.65	1.30	0.31	0.52
Sym1	CC	0.89	1.48	0.57	1.07	0.67	1.14
Sym2	SS	0.64	1.08	0.58	1.19	0.27	0.43
Sym2	CC	0.67	1.23	0.63	1.23	0.41	0.74
Unsym	SS	0.46	0.66	0.66	1.15	0.25	0.49
Unsym	CC	0.48	0.97	0.65	1.21	0.17	0.33
All 60		0.58	1.48	0.57	1.30	0.27	1.14

The deviations depend mainly on the stringer pitch (Figure 3a). For $b_s = 285.5$ mm the GBT mean deviation is 0.25 %. For $b_s = 150$ mm it is 0.91 %, and all 30 GBT results and all 30 strip results lie below Abaqus (GBT −1.48 % to −0.48 %, strip −1.30 % to −0.65 %). This offset is therefore common to the two plate models and not caused by the GBT mode basis; the refinement study of Section 4.6 shows that it is not a discretisation error of GBT either, since finer segments lower the GBT result slightly. Its likely origins are differences between the shell elements and the plate model in the short skin bays of the narrow pitch, and the representation of the tie; it has not been resolved in this study. The largest GBT deviation, −1.48 %, occurs for Sym1, CC, $b_s = 150$ mm, $\beta = 3$ (GBT 12.871, strip 12.972, Abaqus 13.064 N/mm).

4.4. Periodic side edges

The 30 SS cases with periodic side edges are summarised in Table 6 and listed in full in Table A1. GBT agrees with Abaqus within 0.57 % on average and 1.25 % at most (strip: 0.59 % and 1.37 %). As for the symmetry edges, the mean deviation is larger for $b_s = 150$ mm (0.93 %, maximum 1.25 %) than for $b_s = 285.5$ mm

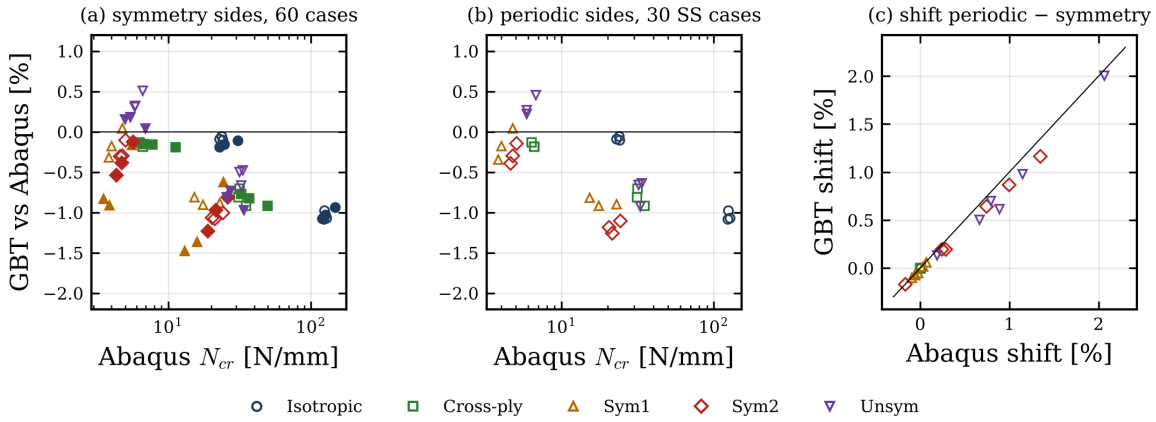


Figure 3. Critical loads of GBT against Abaqus. (a) Deviation for the 60 cases with symmetry side edges (open symbols: SS, filled symbols: CC). (b) Deviation for the 30 SS cases with periodic side edges. (c) Change of the critical load caused by periodic instead of symmetry side edges, $100\,(N_{cr}^{per} - N_{cr}^{sym})/N_{cr}^{sym}$, GBT against Abaqus (30 SS cases; the line marks equality).

(0.20 %, maximum 0.46 %). The periodic formulation is therefore as accurate as the symmetry formulation (Figure 3b).

Table 6. Periodic side edges, SS ends, 30 cases: absolute deviation of the critical load in %, and change of the critical load from symmetry to periodic side edges, $\Delta = 100\,(N_{cr}^{per} - N_{cr}^{sym})/N_{cr}^{sym}$, range over the six cases of each lay-up.

Lay-up	GBT vs Abaqus, mean	max	Strip vs Abaqus, mean	max	Δ Abaqus [%]	Δ GBT [%]	Δ strip [%]
Isotropic	0.56	1.08	0.59	1.10	0.00	0.00	0.00
Cross-ply	0.49	0.91	0.41	0.69	0.00	0.00	0.00
Sym1	0.53	0.92	0.65	1.30	−0.10 to +0.07	−0.10 to +0.06	−0.10 to +0.06
Sym2	0.73	1.25	0.60	1.25	−0.17 to +1.35	−0.17 to +1.16	−0.19 to +1.23
Unsym	0.53	0.92	0.71	1.37	+0.19 to +2.06	+0.14 to +2.01	+0.15 to +2.10
All 30	0.57	1.25	0.59	1.37			

A more demanding test than the absolute critical load is the *change* Δ that periodic instead of symmetry side edges produce, because the two models of each case share their discretisation and the common offset of Section 4.3 largely cancels. GBT reproduces the change found by Abaqus with a mean deviation of 0.05 percentage points (maximum 0.27 points) and a correlation coefficient of 0.995 (Figure 3c); the strip model gives 0.03 points (maximum 0.23) and 0.996.

4.5. Effect of periodic side edges

For the isotropic and the cross-ply panel, periodic and symmetry side edges give the same critical load in all three models (Table 6 and Tables A1 and A2; the largest difference is 0.01 %). The reason is that the critical mode of an orthotropic panel without coupling is symmetric about the mid-lines of the skin bays, so that the symmetry conditions $U_X = \theta = 0$ are satisfied by the exact periodic mode. For coupled laminates this is no longer the case. A symmetry plane at the side edge implies that the neighbouring cell is the mirror image of the cell, and mirroring changes the sign of the ply angles: a cell with the angle-ply laminates Sym1 or Sym2 is mirrored into a cell with the opposite bending–twisting coupling. The symmetry model therefore represents a panel that is not the real one, while the periodic model represents the real, infinitely repeated panel. Bending–twisting and membrane–bending coupling skew the nodal lines of the buckles (Figure 5), and the symmetry conditions force the skewed mode back to zero rotation at the sides.

The size of the effect follows the strength of the coupling (Figure 4). For Sym1 (±45°) the change remains within ±0.1 %, although its bending–twisting coupling is not zero, which indicates that its critical mode is nearly symmetric about the bay mid-lines. For Sym2 (60°/−30°) and Unsym it reaches 1.35 % and 2.06 % in Abaqus with SS ends (GBT: 1.16 % and 2.01 %). With CC ends GBT predicts much larger changes, +5.50 % to +18.16 % for Sym2 and +8.53 % to +11.61 % for Unsym, and the strip model confirms them independently

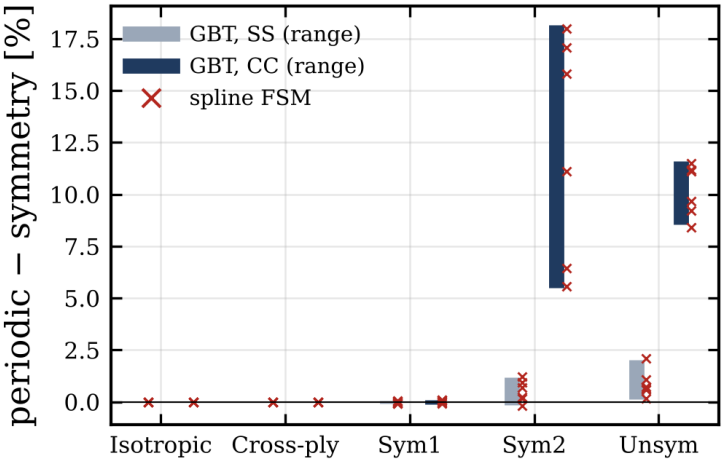


Figure 4. Change of the critical load from symmetry to periodic side edges, $100\,(N_{cr}^{per} - N_{cr}^{sym})/N_{cr}^{sym}$, by lay-up: range of GBT for SS and CC ends (bars, six cases each) and individual results of the spline finite strip model (crosses).

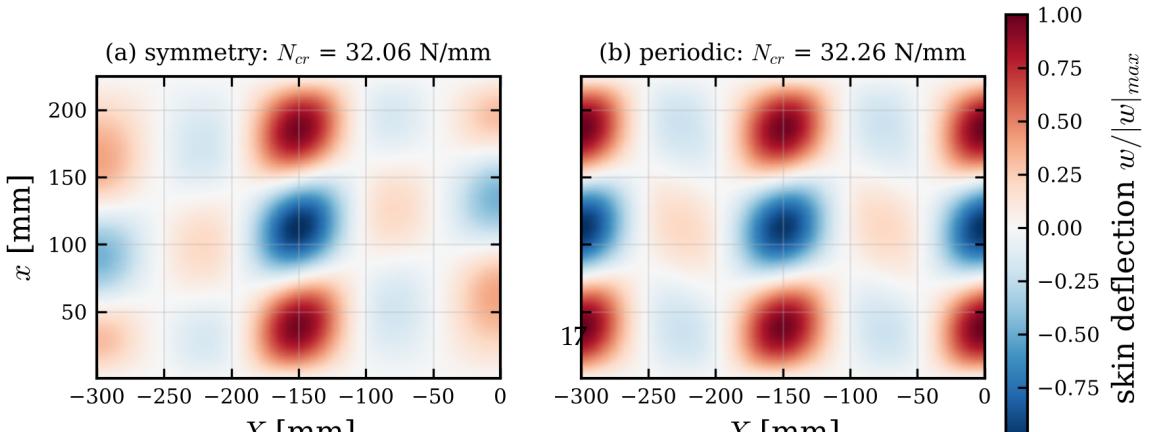
Table 7. Critical load N_{cr} [N/mm] with the default discretisation and with refined spline spans or cross-section segments (symmetry side edges); in brackets the change from the default.

Case	Default	Spans $h_x/2$	Segments $b_s/40$
Isotropic, SS, $b_s = 150$, $\beta = 1.5$	122.674	122.592 (−0.07 %)	122.407 (−0.22 %)
Sym2, SS, $b_s = 150$, $\beta = 3$	20.000	20.002 (+0.01 %)	19.964 (−0.18 %)
Unsym, CC, $b_s = 150$, $\beta = 1.5$	27.002	27.000 (−0.01 %)	26.881 (−0.45 %)
Sym1, CC, $b_s = 285.5$, $\beta = 3$	3.4722	3.4789 (+0.19 %)	3.4703 (−0.05 %)
Sym1, CC, $b_s = 285.5$, $\beta = 1.5$	3.8165	3.8266 (+0.27 %)	3.8133 (−0.08 %)

For the last case, a further halving of the spans ($h_x/4$) gives 3.8348 N/mm (+0.48 %), so that the spline discretisation converges from below in this case; the remaining uncertainty is about 0.2 %.

4.7. Buckling modes and modal participation

Figure 5 shows the skin deflection of the critical mode of the Unsym panel with SS ends, $b_s = 150$ mm and $\beta = 1.5$, for the two side conditions. In both cases the mode is a local skin buckle with three half-waves along the length; the buckles are skewed by the coupling. With symmetry side edges (Figure 5a) the half-bays at the sides are forced to zero slope across the edge and their deflection differs from that of the central bay; with periodic side edges (Figure 5b) the deflection at $X = -W$ and $X = 0$ is identical (largest difference 2.2×10^{-16} of the maximum, against 0.72 with symmetry edges), and the half-bays at the two sides join into one full bay of the same shape as the central one. The critical load changes from 32.06 to 32.26 N/mm (+0.62 %).



4.8. Computational cost

With the full basis of 482–569 modes, a GBT analysis (cross-section analysis, pre-buckling solution and eigenvalue problem) takes 8.4 s on average (3.1–21.3 s) on two cores of a standard workstation, against 7.3 s for the spline finite strip model with twice as many unknowns. The cross-section analysis is independent of the panel length, and the sparse factors of the geometric matrices (Section 3.10) are reused for all lengths of a parametric study.

4.9. Limitations

The following limitations should be noted. (i) GBT and the strip model share the plate assumptions: laminated Reissner–Mindlin walls with a shear correction factor of $5/6$, an offset tie represented by stiff link walls or rigid links, and linear buckling about a first-order pre-buckling state. An error common to both, such as the -1 % offset for $b_s = 150$ mm, cannot be detected by comparing them. (ii) The link walls are a penalty idealisation of the tie; the results are insensitive to the penalty factor (122.656, 122.663 and 122.660 N/mm for $k_L = 10^2$, 10^3 and 10^4, Isotropic, SS, $b_s = 150$ mm, $\beta = 1.5$), but a stiff twisting stiffness D_{66} of the link walls would overestimate the critical load by 23 %, which is why the bending and twisting stiffnesses of the link walls are reduced. (iii) The periodic condition of Eq. (10) keeps the width of the cell fixed. A periodic condition with free lateral expansion, in which the two side edges differ by a common jump in U_X and u, is not yet implemented in GBT. (iv) A cell of width $2b_s$ admits the Bloch wavenumbers $\kappa b_s = 0$ and π only; buckling modes of an infinite panel with a longer period across the stringers require a wider cell or complex Bloch modes. (v) The verified model uses the full basis of 482–569 modes; a reduced, hierarchical basis of a few tens of modes, as typically used in GBT, has not yet been validated for the coupled lay-ups with clamped ends. (vi) The rotation β_x of the Mindlin modes does not enter the load term, which is consistent with the shell edge load of the reference model.

5. Conclusions

A shear-deformable Generalized Beam Theory with exact periodic boundary conditions was developed for the buckling analysis of omega-stringer-stiffened composite panels. The main conclusions are:

1. Periodic side edges can be built into every family of GBT deformation modes. The conventional modes are obtained by projecting out constraint rows family by family, the warping of the global and distortional modes by a constrained Vlasov least-squares problem, and the shear and Mindlin modes by merging the nodal functions of the two side edges. Every mode, and hence any truncated basis, satisfies the periodic condition to machine precision. The rigid rotation of the cross-section is eliminated, and three additional distortional modes arise from the coupled lateral motion and rotation of the side edges.
2. The formulation retains the structure and the symbols of classical GBT. The modal matrices $\bar{\mathbf{B}}$, $\bar{\mathbf{C}}$, $\bar{\mathbf{D}}_1$, $\bar{\mathbf{D}}_2$, $\bar{\mathbf{F}}_1$ and $\bar{\mathbf{H}}_1$ were given explicitly for laminated walls with membrane–bending and anisotropic coupling, and the system reduces exactly to the classical GBT buckling equations and the classical geometric matrices for uncoupled orthotropic walls under uniform compression.
3. Against 90 Abaqus shell models, the critical loads agree within 0.58 % on average (maximum 1.48 %) with symmetry side edges and within 0.57 % (maximum 1.25 %) with periodic side edges. An independent spline finite strip model agrees with GBT within 0.27 % on average. The change caused by periodic instead of symmetry side edges is reproduced with a mean deviation of 0.05 percentage points and a correlation of 0.995.
4. For uncoupled laminates the symmetry and periodic models are equivalent. For laminates with bending–twisting or membrane–bending coupling, the symmetry model represents a mirrored panel and underestimates the critical load of the periodic panel by up to about 2 % for simply supported and, according to GBT and the strip model, by 5–18 % for clamped loaded edges.
5. GBT identifies all critical modes of the study as local skin buckling with a minor participation of global and distortional modes, at a computing time of a few seconds per case.

Future work will address the confirmation of the clamped periodic cases with shell finite element models, the periodic condition with free lateral expansion, reduced hierarchical mode bases for parametric studies and surrogate models, and the comparison with the closed-form periodic solutions of El Yaakoubi-Mesbah and Mittelstedt [17].

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preparation, A.M.E.; writing—review and editing, N.K. and C.M.; supervision, N.K. and C.M.; project administration, C.M.; funding acquisition, C.M. All authors have read and agreed to the published version of the manuscript. [To be confirmed by the authors.]

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Appendix A. Critical loads of all cases

Table A1. Critical loads N_{cr} [N/mm], SS ends: symmetry and periodic side edges. Deviations e [%] of GBT and of the spline finite strip model (FSM) from Abaqus; change Δ [%] from symmetry to periodic side edges.

Lay-up	b_s	β	Abaqus sym.	GBT sym.	FSM sym.	e_{GBT} sym.	Abaqus per.	GBT per.	FSM per.	e_{GBT} per.	e_{FSM} per.	Δ Abaqus	Δ GBT
Isotropic	150	0.75	127.650	126.284	126.345	−1.07	127.650	126.284	126.345	−1.07	−1.02	+0.00	+0.00
Isotropic	150	1.5	123.880	122.674	122.617	−0.97	123.880	122.674	122.617	−0.97	−1.02	+0.00	+0.00
Isotropic	150	3	122.740	121.412	121.395	−1.08	122.740	121.412	121.395	−1.08	−1.10	+0.00	+0.00
Isotropi2	285.5	0.75	23.979	23.954	24.022	−0.10	23.979	23.954	24.022	−0.10	+0.18	+0.00	+0.00
Isotropi2	285.5	1.5	23.720	23.707	23.742	−0.05	23.720	23.707	23.742	−0.05	+0.09	+0.00	+0.00
Isotropi2	285.5	3	22.873	22.852	22.898	−0.09	22.873	22.852	22.898	−0.09	+0.11	+0.00	+0.00
Cross-ply	150	0.75	34.848	34.530	34.606	−0.91	34.848	34.530	34.606	−0.91	−0.69	+0.00	+0.00
Cross-ply	150	1.5	31.464	31.243	31.258	−0.70	31.464	31.243	31.258	−0.70	−0.65	+0.00	+0.00
Cross-ply	150	3	31.061	30.810	30.846	−0.81	31.061	30.810	30.846	−0.81	−0.69	+0.00	+0.00
Cross-ply	285.5	0.75	6.607	6.595	6.618	−0.18	6.607	6.595	6.618	−0.18	+0.17	+0.00	+0.00
Cross-ply	285.5	1.5	6.609	6.597	6.620	−0.18	6.609	6.597	6.620	−0.18	+0.17	+0.00	+0.00
Cross-ply	285.5	3	6.304	6.296	6.310	−0.13	6.304	6.296	6.310	−0.13	+0.10	+0.00	+0.00
Sym1	150	0.75	22.894	22.692	22.597	−0.88	22.880	22.675	22.583	−0.90	−1.30	−0.06	−0.07
Sym1	150	1.5	17.470	17.312	17.268	−0.90	17.475	17.315	17.272	−0.92	−1.16	+0.03	+0.02
Sym1	150	3	15.192	15.069	15.078	−0.81	15.202	15.078	15.087	−0.82	−0.76	+0.07	+0.06
Sym1	285.5	0.75	4.763	4.765	4.776	+0.04	4.760	4.762	4.772	+0.04	+0.25	−0.06	−0.06
Sym1	285.5	1.5	4.015	4.008	4.024	−0.17	4.011	4.004	4.020	−0.17	+0.22	−0.10	−0.10
Sym1	285.5	3	3.833	3.821	3.841	−0.31	3.832	3.819	3.840	−0.34	+0.21	−0.03	−0.05
Sym2	150	0.75	24.112	23.870	23.824	−1.00	24.291	24.024	23.988	−1.10	−1.25	+0.74	+0.65
Sym2	150	1.5	21.102	20.875	20.888	−1.08	21.386	21.118	21.144	−1.25	−1.13	+1.35	+1.16
Sym2	150	3	20.214	20.000	20.044	−1.06	20.415	20.174	20.229	−1.18	−0.91	+0.99	+0.87
Sym2	285.5	0.75	5.021	5.016	5.032	−0.10	5.033	5.026	5.041	−0.14	+0.16	+0.24	+0.20
Sym2	285.5	1.5	4.766	4.752	4.771	−0.29	4.758	4.744	4.762	−0.29	+0.08	−0.17	−0.17
Sym2	285.5	3	4.607	4.593	4.613	−0.30	4.620	4.602	4.623	−0.39	+0.06	+0.28	+0.20
Unsym	150	0.75	33.269	33.110	32.972	−0.48	33.489	33.277	33.159	−0.63	−0.99	+0.66	+0.50
Unsym	150	1.5	32.275	32.062	31.905	−0.66	32.561	32.260	32.115	−0.92	−1.37	+0.89	+0.62
Unsym	150	3	31.464	31.309	31.177	−0.49	31.825	31.616	31.511	−0.66	−0.99	+1.15	+0.98
Unsym	285.5	0.75	6.638	6.672	6.665	+0.51	6.775	6.806	6.805	+0.46	+0.44	+2.06	+2.01
Unsym	285.5	1.5	5.869	5.888	5.887	+0.32	5.880	5.896	5.896	+0.27	+0.27	+0.19	+0.14
Unsym	285.5	3	5.818	5.836	5.835	+0.31	5.864	5.877	5.877	+0.22	+0.22	+0.79	+0.70

Table A2. Critical loads N_{cr} [N/mm], CC ends. Symmetry side edges: Abaqus, GBT and FSM; periodic side edges: GBT and FSM (no Abaqus results). Deviation e [%] of GBT from Abaqus; change Δ [%] from symmetry to periodic side edges.

Lay-up	b_s	β	Abaqus sym.	GBT sym.	FSM sym.	e_{GBT} sym.	GBT per.	FSM per.	Δ GBT	Δ FSM
Isotropic	150	0.75	146.600	145.225	145.165	−0.94	145.225	145.165	+0.00	+0.00
Isotropic	150	1.5	126.100	124.801	124.834	−1.03	124.801	124.834	+0.00	+0.00
Isotropic	150	3	119.420	118.133	118.166	−1.08	118.133	118.166	+0.00	+0.00
Isotropic	285.5	0.75	30.678	30.645	30.694	−0.11	30.645	30.694	+0.00	+0.00
Isotropic	285.5	1.5	24.713	24.675	24.736	−0.15	24.675	24.736	+0.00	+0.00
Isotropic	285.5	3	22.913	22.870	22.935	−0.19	22.870	22.935	+0.00	+0.00
Cross-ply	150	0.75	49.330	48.878	48.899	−0.92	48.878	48.899	+0.00	+0.00
Cross-ply	150	1.5	36.806	36.503	36.538	−0.82	36.503	36.538	+0.00	+0.00
Cross-ply	150	3	32.275	32.027	32.052	−0.77	32.027	32.052	+0.00	+0.00
Cross-ply	285.5	0.75	11.250	11.229	11.243	−0.19	11.228	11.243	−0.01	+0.00
Cross-ply	285.5	1.5	7.749	7.737	7.754	−0.15	7.737	7.754	+0.00	+0.00
Cross-ply	285.5	3	6.709	6.699	6.716	−0.15	6.699	6.716	+0.00	+0.00
Sym1	150	0.75	24.185	24.034	23.947	−0.62	24.047	23.959	+0.05	+0.05
Sym1	150	1.5	15.810	15.595	15.641	−1.36	15.606	15.654	+0.07	+0.08
Sym1	150	3	13.064	12.871	12.972	−1.48	12.882	12.984	+0.09	+0.09
Sym1	285.5	0.75	5.573	5.564	5.581	−0.16	5.564	5.580	+0.00	−0.02
Sym1	285.5	1.5	3.852	3.817	3.861	−0.91	3.813	3.858	−0.10	−0.08
Sym1	285.5	3	3.501	3.472	3.511	−0.83	3.471	3.510	−0.03	−0.03
Sym2	150	0.75	26.063	25.852	25.743	−0.81	30.252	30.135	+17.02	+17.06
Sym2	150	1.5	21.468	21.260	21.224	−0.97	23.563	23.579	+10.83	+11.10
Sym2	150	3	18.880	18.648	18.709	−1.23	21.619	21.668	+15.93	+15.82
Sym2	285.5	0.75	5.663	5.656	5.669	−0.12	6.683	6.688	+18.16	+17.97
Sym2	285.5	1.5	4.716	4.698	4.726	−0.38	5.005	5.030	+6.53	+6.43
Sym2	285.5	3	4.317	4.294	4.326	−0.53	4.530	4.567	+5.50	+5.57
Unsym	150	0.75	33.614	33.288	33.207	−0.97	36.129	35.997	+8.53	+8.40
Unsym	150	1.5	27.201	27.002	26.913	−0.73	30.029	29.924	+11.21	+11.19
Unsym	150	3	25.558	25.351	25.281	−0.81	27.821	27.722	+9.74	+9.66
Unsym	285.5	0.75	6.912	6.915	6.921	+0.04	7.718	7.717	+11.61	+11.50
Unsym	285.5	1.5	5.401	5.411	5.412	+0.19	6.013	6.014	+11.13	+11.12
Unsym	285.5	3	4.986	4.994	4.997	+0.16	5.455	5.458	+9.23	+9.23

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